

Tight-Binding Hamiltonian

The real-space matrix elements define

$$H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} H(\mathbf{R}), \quad (1)$$

where $\mathbf{k}$ is given in reduced coordinates and $H_{mn}(\mathbf{R}) = \langle \mathbf{0}m | \hat{H} | \mathbf{R}n \rangle$.
The direct-lattice basis vectors used in $\mathbf{R} = R_1\mathbf{a}_1 + R_2\mathbf{a}_2 + R_3\mathbf{a}_3$ are

$$\mathbf{a}_1 = (3.01435, -5.221, 0) \quad (2)$$

$$\mathbf{a}_2 = (3.01435, 5.221, 0) \quad (3)$$

$$\mathbf{a}_3 = (0, 0, 23) \quad (4)$$

The selected contributions up to the requested neighbor shell are

$$H(\mathbf{k}) \approx \sum_{\text{listed}} t_{mn}(\mathbf{R}) e^{i\mathbf{k}\cdot\mathbf{R}} |m\rangle\langle n| + h.c., \quad (5)$$

with $t_{mn}(\mathbf{R}) = H_{mn}(\mathbf{R})$ in eV:

Second-quantization form

The Hamiltonian in second-quantization reads

$$H = \sum_{\mathbf{k}} \sum_{m,n} H_{mn}(\mathbf{k}) c_{m\mathbf{k}}^\dagger c_{n\mathbf{k}}, \quad (6)$$

where $H_{mn}(\mathbf{k}) = \sum_{\mathbf{R}} H_{mn}(\mathbf{R}) e^{i\mathbf{k}\cdot\mathbf{R}}$.

Expanded term by term (orbital labels: P1: d_{xz} , P1: d_{yz} , P2: d_{xz} , P2: d_{yz}):

$$H = \sum_{\mathbf{k}} \left[\varepsilon_{\text{P1:}d_{xz}} c_{\text{P1:}d_{xz},\mathbf{k}}^\dagger c_{\text{P1:}d_{xz},\mathbf{k}} + \varepsilon_{\text{P1:}d_{yz}} c_{\text{P1:}d_{yz},\mathbf{k}}^\dagger c_{\text{P1:}d_{yz},\mathbf{k}} \right. \quad (7)$$

$$\left. + \varepsilon_{\text{P2:}d_{xz}} c_{\text{P2:}d_{xz},\mathbf{k}}^\dagger c_{\text{P2:}d_{xz},\mathbf{k}} + \varepsilon_{\text{P2:}d_{yz}} c_{\text{P2:}d_{yz},\mathbf{k}}^\dagger c_{\text{P2:}d_{yz},\mathbf{k}} \right] \quad (8)$$

$$+ f_{\text{P1:}d_{xz} \rightarrow \text{P1:}d_{yz}}(\mathbf{k}) c_{\text{P1:}d_{xz},\mathbf{k}}^\dagger c_{\text{P1:}d_{yz},\mathbf{k}} + f_{\text{P1:}d_{xz} \rightarrow \text{P1:}d_{yz}}^*(\mathbf{k}) c_{\text{P1:}d_{yz},\mathbf{k}}^\dagger c_{\text{P1:}d_{xz},\mathbf{k}} \quad (9)$$

$$+ f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{xz}}(\mathbf{k}) c_{\text{P1:}d_{xz},\mathbf{k}}^\dagger c_{\text{P2:}d_{xz},\mathbf{k}} + f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{xz}}^*(\mathbf{k}) c_{\text{P2:}d_{xz},\mathbf{k}}^\dagger c_{\text{P1:}d_{xz},\mathbf{k}} \quad (10)$$

$$+ f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) c_{\text{P1:}d_{xz},\mathbf{k}}^\dagger c_{\text{P2:}d_{yz},\mathbf{k}} + f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{yz}}^*(\mathbf{k}) c_{\text{P2:}d_{yz},\mathbf{k}}^\dagger c_{\text{P1:}d_{xz},\mathbf{k}} \quad (11)$$

$$+ f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{xz}}(\mathbf{k}) c_{\text{P1:}d_{yz},\mathbf{k}}^\dagger c_{\text{P2:}d_{xz},\mathbf{k}} + f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{xz}}^*(\mathbf{k}) c_{\text{P2:}d_{xz},\mathbf{k}}^\dagger c_{\text{P1:}d_{yz},\mathbf{k}} \quad (12)$$

$$+ f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) c_{\text{P1:}d_{yz},\mathbf{k}}^\dagger c_{\text{P2:}d_{yz},\mathbf{k}} + f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{yz}}^*(\mathbf{k}) c_{\text{P2:}d_{yz},\mathbf{k}}^\dagger c_{\text{P1:}d_{yz},\mathbf{k}} \quad (13)$$

$$\left. + f_{\text{P2:}d_{xz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) c_{\text{P2:}d_{xz},\mathbf{k}}^\dagger c_{\text{P2:}d_{yz},\mathbf{k}} + f_{\text{P2:}d_{xz} \rightarrow \text{P2:}d_{yz}}^*(\mathbf{k}) c_{\text{P2:}d_{yz},\mathbf{k}}^\dagger c_{\text{P2:}d_{xz},\mathbf{k}} \right], \quad (14)$$

where the on-site energies are

$$\varepsilon_{\text{P1:}d_{xz}} = \text{Re}[H_{\text{P1:}d_{xz}, \text{P1:}d_{xz}}(\mathbf{0})] = -4.04203 \text{ eV} \quad (15)$$

$$\varepsilon_{\text{P1:}d_{yz}} = \text{Re}[H_{\text{P1:}d_{yz}, \text{P1:}d_{yz}}(\mathbf{0})] = -4.04262 \text{ eV} \quad (16)$$

$$\varepsilon_{\text{P2:}d_{xz}} = \text{Re}[H_{\text{P2:}d_{xz}, \text{P2:}d_{xz}}(\mathbf{0})] = -4.04203 \text{ eV} \quad (17)$$

$$\varepsilon_{\text{P2:}d_{yz}} = \text{Re}[H_{\text{P2:}d_{yz}, \text{P2:}d_{yz}}(\mathbf{0})] = -4.04262 \text{ eV} \quad (18)$$

The structure functions are defined as

$$f_{\text{P1:}d_{xz} \rightarrow \text{P1:}d_{yz}}(\mathbf{k}) = -0.015339 \quad (19)$$

$$f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{xz}}(\mathbf{k}) = -0.082235 + 0.037878 e^{i\mathbf{k} \cdot (\mathbf{a}_1)} + 0.032117 e^{i\mathbf{k} \cdot (-\mathbf{a}_2)} \quad (20)$$

$$f_{\text{P1:}d_{xz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) = 0.168186 e^{i\mathbf{k} \cdot (-\mathbf{a}_2)} - 0.024001 - 0.022112 e^{i\mathbf{k} \cdot (\mathbf{a}_1)} \quad (21)$$

$$f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{xz}}(\mathbf{k}) = 0.168186 e^{i\mathbf{k} \cdot (-\mathbf{a}_2)} - 0.024001 - 0.022112 e^{i\mathbf{k} \cdot (\mathbf{a}_1)} \quad (22)$$

$$f_{\text{P1:}d_{yz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) = -0.083117 e^{i\mathbf{k} \cdot (\mathbf{a}_1)} + 0.040589 e^{i\mathbf{k} \cdot (-\mathbf{a}_2)} + 0.034953 \quad (23)$$

$$f_{\text{P2:}d_{xz} \rightarrow \text{P2:}d_{yz}}(\mathbf{k}) = -0.015339 \quad (24)$$

Table 1: Hopping parameters extracted from `wannier90_hr.dat`.

from	to	NN	distance ($\text{\AA}$)	R_1	R_2	R_3	$\text{Re}[t]$ (eV)	$\text{Im}[t]$ (eV)	$ t $ (eV)
P1: d_{yz}	P1: d_{yz}	0	0.000000	0	0	0	-4.042 617	0.000 000	4.042 617
P2: d_{yz}	P2: d_{yz}	0	0.000000	0	0	0	-4.042 617	0.000 000	4.042 617
P1: d_{xz}	P1: d_{xz}	0	0.000000	0	0	0	-4.042 027	0.000 000	4.042 027
P2: d_{xz}	P2: d_{xz}	0	0.000000	0	0	0	-4.042 027	0.000 000	4.042 027
P1: d_{xz}	P1: d_{yz}	0	0.000000	0	0	0	-0.015 339	0.000 000	0.015 339
P1: d_{yz}	P1: d_{xz}	0	0.000000	0	0	0	-0.015 339	0.000 000	0.015 339
P2: d_{xz}	P2: d_{yz}	0	0.000000	0	0	0	-0.015 339	0.000 000	0.015 339
P2: d_{yz}	P2: d_{xz}	0	0.000000	0	0	0	-0.015 339	0.000 000	0.015 339
P1: d_{xz}	P2: d_{yz}	1	3.480667	0	-1	0	0.168 186	0.000 000	0.168 186
P1: d_{yz}	P2: d_{xz}	1	3.480667	0	-1	0	0.168 186	0.000 000	0.168 186
P2: d_{xz}	P1: d_{yz}	1	3.480667	0	1	0	0.168 186	0.000 000	0.168 186
P2: d_{yz}	P1: d_{xz}	1	3.480667	0	1	0	0.168 186	0.000 000	0.168 186
P1: d_{yz}	P2: d_{yz}	1	3.480667	1	0	0	-0.083 117	0.000 000	0.083 117
P2: d_{yz}	P1: d_{yz}	1	3.480667	-1	0	0	-0.083 117	0.000 000	0.083 117
P1: d_{xz}	P2: d_{xz}	1	3.480667	0	0	0	-0.082 235	0.000 000	0.082 235
P2: d_{xz}	P1: d_{xz}	1	3.480667	0	0	0	-0.082 235	0.000 000	0.082 235
P1: d_{yz}	P2: d_{yz}	1	3.480667	0	-1	0	0.040 589	0.000 000	0.040 589
P2: d_{yz}	P1: d_{yz}	1	3.480667	0	1	0	0.040 589	0.000 000	0.040 589
P1: d_{xz}	P2: d_{xz}	1	3.480667	1	0	0	0.037 878	0.000 000	0.037 878
P2: d_{xz}	P1: d_{xz}	1	3.480667	-1	0	0	0.037 878	0.000 000	0.037 878
P1: d_{yz}	P2: d_{yz}	1	3.480667	0	0	0	0.034 953	0.000 000	0.034 953
P2: d_{yz}	P1: d_{yz}	1	3.480667	0	0	0	0.034 953	0.000 000	0.034 953
P1: d_{xz}	P2: d_{xz}	1	3.480667	0	-1	0	0.032 117	0.000 000	0.032 117
P2: d_{xz}	P1: d_{xz}	1	3.480667	0	1	0	0.032 117	0.000 000	0.032 117
P1: d_{xz}	P2: d_{yz}	1	3.480667	0	0	0	-0.024 001	0.000 000	0.024 001
P1: d_{yz}	P2: d_{xz}	1	3.480667	0	0	0	-0.024 001	0.000 000	0.024 001
P2: d_{xz}	P1: d_{yz}	1	3.480667	0	0	0	-0.024 001	0.000 000	0.024 001
P2: d_{yz}	P1: d_{xz}	1	3.480667	0	0	0	-0.024 001	0.000 000	0.024 001
P1: d_{xz}	P2: d_{yz}	1	3.480667	1	0	0	-0.022 112	0.000 000	0.022 112
P1: d_{yz}	P2: d_{xz}	1	3.480667	1	0	0	-0.022 112	0.000 000	0.022 112
P2: d_{xz}	P1: d_{yz}	1	3.480667	-1	0	0	-0.022 112	0.000 000	0.022 112
P2: d_{yz}	P1: d_{xz}	1	3.480667	-1	0	0	-0.022 112	0.000 000	0.022 112